Lending Under Imperfect Information

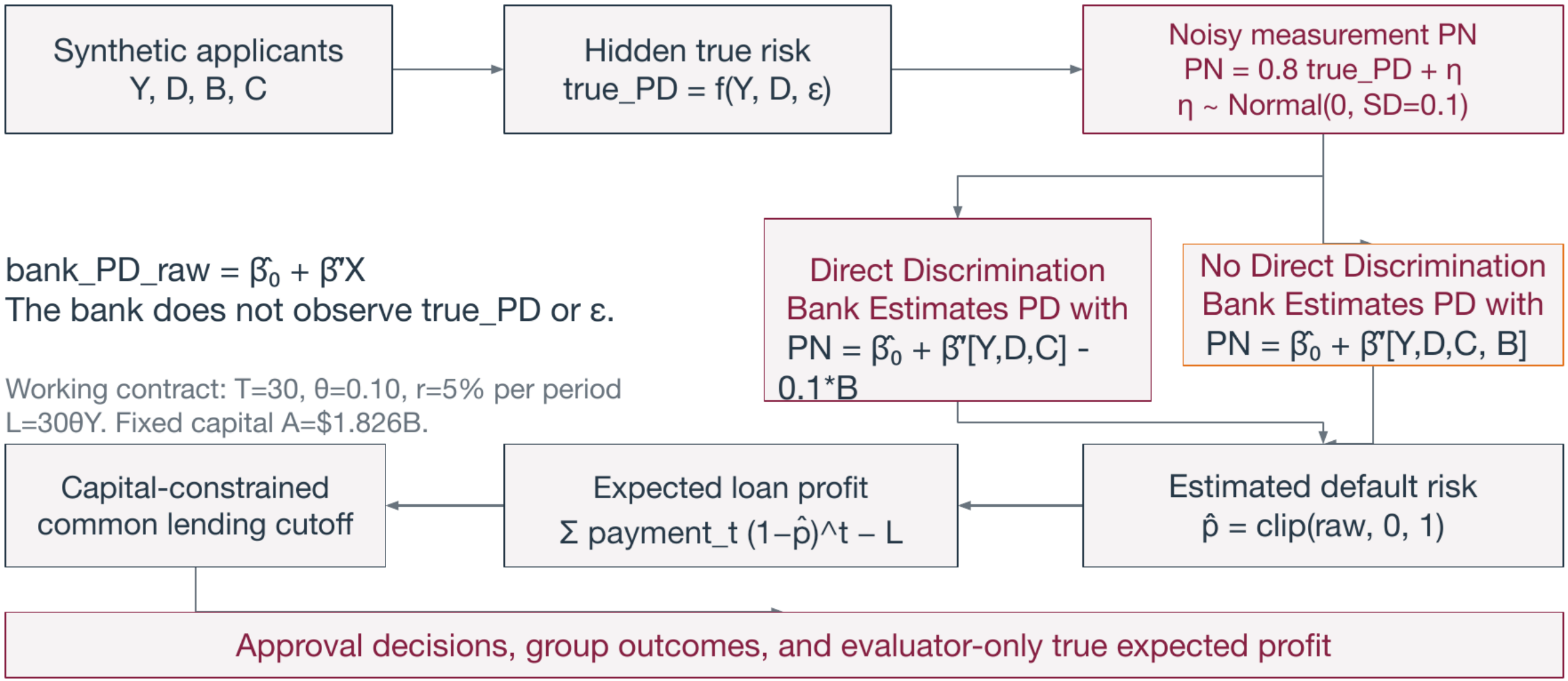
Follow-up: group effects, clipping and lending order

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Which lending differences come from the fitted score, and which come from clipping and the lending rule?

20,000 evaluation applicants. Contracts and capital stay fixed.

From True Risk to Lending Decisions



Source: professor-model Steps 2–7B, saved specifications and evaluation artifacts.

The Revised Direct and No-Direct Comparison

PN = 0.8 true_PD + η remains the regression target

Scenario	Raw bank score	Interpretation
No manual adjustment Aware model	f_W(Y, D, C, B)	B remains a fitted predictor
Direct adjustment Blind base	f_F(Y, D, C) – 0.1 B	Lower estimated PD favors B=1
Matched control Blind base	f_F(Y, D, C)	Same fit without the adjustment

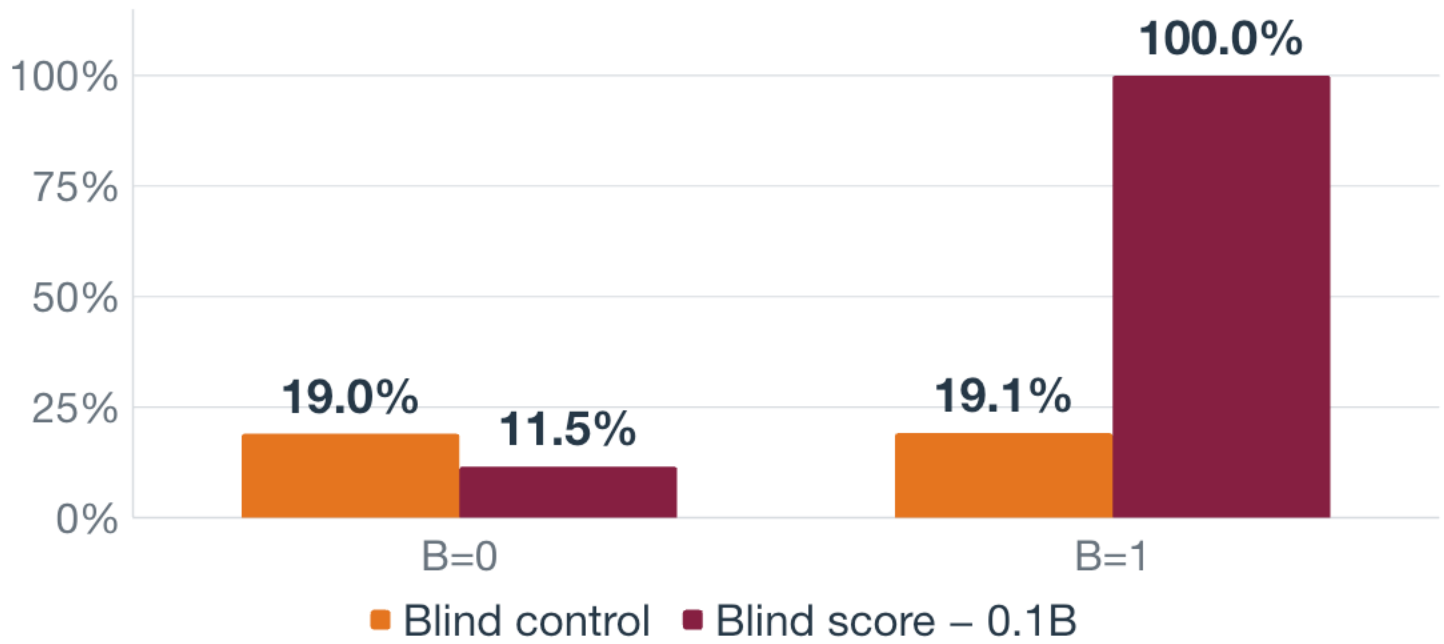
The direct/control comparison keeps the blind fit fixed

Comparing aware and blind fits also changes the intercept and other slopes.

Source: revised_lending/portfolio_comparison.csv and the supplied reference slide 2. Canonical synthetic fits.

The –0.1B Adjustment Favors B=1

Approval rate by group



B=1 minus B=0 approval gap

+88.45 pp

Blind control +0.16 pp

True profit versus control

–$7.332M

All 3,030 B=1 requests receive funding after their estimated PD clips to zero.

Source: revised_lending/group_outcome_audit.csv and portfolio_comparison.csv. One canonical realization.

Moving the Tie Leaves Lending Unchanged

Replace only exact zeros with a predeclared positive floor

Quantity	Original direct policy	Positive-floor sensitivity
Shared tie score	0%	0.013924%
Applicants in the tie	3,764	3,764
Total approvals	4,990	4,990
True portfolio profit	$2.3593B	$2.3593B

0 approval decisions change

Perceived profit changes by –$7.855M. The tie remains intact.

Source: mechanism_audit/positive_floor_derivation.json and portfolio_comparison.csv. Canonical sensitivity.

Removing B’s Contribution Without Refitting

raw_Z = raw_W – saved_beta_B × B

W keeps B. Z removes only its contribution. F is the full blind refit.

Approval gap, B=1 minus B=0 (pp)



Z minus W true profit

+$853,049

Z minus F true profit

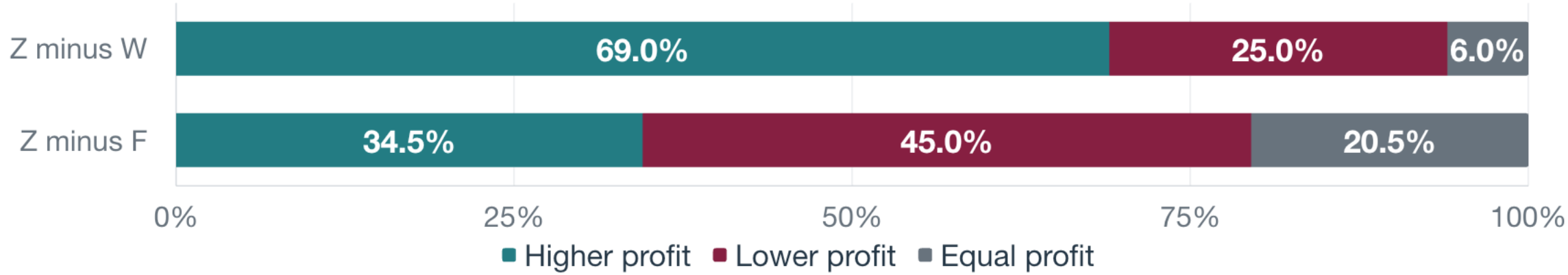
+$327,762

Z and F have equal group counts but exchange one approved loan.

Source: fixed_coefficient/cd7f3ee268cfc4d0/canonical_policies.csv. One canonical realization.

No Universal Profit Gain Across 200 Draws

Share of draws by signed true-profit difference



Comparison	Mean change	Median change	MCSE of mean
Z minus W	+$12.82M	+$0.593M	$20.53M
Z minus F	–$11.80M	$0	$26.50M

Rare reject-all disagreements dominate the means

Source: fixed_coefficient/cd7f3ee268cfc4d0/noise_summary.csv. Same population and 200 paired saved noise draws.

Ranking Scores and Accounting Probabilities

Role	Original bounded policy	Raw-order sensitivity
Lending order	clip(raw, 0, 1)	raw score
Cash-flow PD	clip(raw, 0, 1)	Same clip(raw, 0, 1)
Indivisible ties	Exact bounded scores	Exact raw scores

Draw 008, aware: the zero-score block requires 110.01% of capital.

Raw ordering permits a feasible prefix within the clipped block

Every evaluation request is perceived as profitable, so capital limits lending.
Raw scores rank requests. Cash flows use the same bounded PD.

Source: raw_ordering.yaml and saved raw-ordering bank freezes. Fixed probabilities, contracts and capital.

Nine Reject-All Policies Now Fund Loans

Approvals shown as original / raw ordering

Case	W: aware	Z: B removed	F: blind refit
Canonical	3,900 / 3,900	3,802 / 3,802	3,802 / 3,802
Draw 008	0 / 5,107	0 / 4,741	4,744 / 4,744
Draw 064	0 / 3,898	3,701 / 3,701	0 / 3,701
Draw 089	4,331 / 4,331	0 / 4,163	4,161 / 4,161
Draw 092	0 / 4,222	0 / 3,865	3,861 / 3,861
Draw 154	0 / 4,812	3,454 / 3,454	0 / 3,454

Risk estimates and per-request profits stay unchanged

The aware approval gap in draw 154 remains +88.25 pp.

Source: raw_ordering/d0421da2ea691f2e/policies.csv. Six selected diagnostics, not a random uncertainty sample.

Decisions Before the Final Research Report

Lending order	May raw scores rank requests while bounded PD determines profits?
Contract interpretation	Is 5% per period with the 30-period growing schedule intended?
Measurement noise	Keep SD 0.1 as the working case, or predeclare a separate noise study?
Paper scope	Should the final paper add an ML benchmark for this model?

Next deliverable: a final analysis protocol and paper outline